



Emerging trends in India's automotive industry

India's automotive future hinges on two potential developments. First, there's the evolution of Indian-made vehicles into more software-defined models, leveraging the country's robust 4G network for enhanced technological appeal. Alternatively, the government may relax import restrictions, leading to an influx of foreign vehicles from global markets and significantly diversifying the range of options available to Indian consumers. Key points to consider are:

- **Growth of software-defined vehicles.** A potential increase in Indian-produced vehicles incorporating advanced software and technology.
- **Relaxation of import regulations.** Possible easing of tariffs and barriers, leading to a rise in foreign vehicle imports.
- **Market diversification.** An influx of Chinese, American, and European vehicles, offering more options to Indian consumers.

This scenario suggests a significant growth trajectory for the Indian automotive sector. While immediate changes might not be readily apparent, the landscape is expected to undergo a dramatic transformation over the next 5-10 years. Features currently regarded as premium could become standard in new vehicles within the Indian market, heralding a new era marked by enhanced technological integration and a broader diversity in automotive options.

facial recognition camera, similar to iPhone's Face ID, is integrated into the car. Each user could have a personalised account linked to their payment methods, ensuring that all transactions are secure and authorised.

Mercedes-Benz is at the forefront of this innovation, partnering with MasterCard in Germany to trial in-vehicle payments using biometric authentication. This system allows for automatic payments at partnered fuel stations, utilising the vehicle as a virtual credit card.

Apart from fuel and charging payments, this technology extends to other areas like paying for parking or even grabbing a coffee. The versatility of in-vehicle payments signifies a shift towards more integrated and convenient automotive experiences.

Additionally, some automakers are combining mobile apps with in-vehicle systems to streamline this process. The app, linked to the vehicle, acts as an extension, ensuring secure transactions. Others, like BMW, are exploring subscription-based models within their vehicles, allowing customers to subscribe or buy upgrades, such as a heated steering wheel or enhanced connectivity features. This concept,

often referred to as 'hardware as a service,' represents a significant advancement in how we interact with and utilise our vehicles.

Driving made easy with your personal assistant

In the realm of automotive technology, the integration of digital assistants within vehicles is not a new concept. For the past decade, renowned manufacturers like Mercedes and BMW have been embedding voice control systems into their cars. Initially, these systems operated on a set of pre-defined phrases and commands. This rudimentary form of interaction often led to a frustrating experience for users. The primary challenge was the need for precise wording; any deviation from the exact keywords or phrases would result in the vehicle failing to comprehend the user's request. Such limitations highlighted the need for a more intuitive and user-friendly approach to voice interaction in vehicles.

Recent advancements in AI and deep learning models have transformed this domain. The key development is the integration of natural language processing (NLP) within vehicular systems, enabling a more fluid and natural conversation be-

tween the vehicle and its occupants. Unlike the past, users no longer need to memorise specific commands to control basic functions like adjusting the volume, managing seat heaters, or making phone calls. Modern systems, powered by AI, are capable of understanding and processing conversational language. This improvement is further enhanced by the vehicle's ability to connect to cloud services through 4G or 5G networks, allowing for more complex data processing and a smoother user experience.

Moreover, AI-driven digital assistants in vehicles are branching out beyond mere voice commands. Qualcomm, for instance, has showcased the potential for AI to offer personalised recommendations based on the driver's habits and preferences. Imagine a scenario where the AI recognises your routine, such as driving to work every Tuesday at 8 AM. It could automatically set your work location in the navigation system and even suggest actions based on your habits, like ordering your regular coffee in advance. When you arrive at the coffee shop, your order is ready, streamlining your morning routine. These advancements not only enhance the convenience factor but also mark a significant shift towards a software-centric experience in vehicles. The future of automotive technology, heavily influenced by AI and digital assistants, promises a more intelligent, responsive, and user-centric approach, redefining our interaction with our vehicles.

How vehicles talk to each other

One development that has been making waves in the automotive technology sector is the vehicle-to-vehicle (V2V) communication technology. While the concept itself has been around since the 1990s,